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EVENT AND DATE

LECTURE 5 – OPPORTUNITY DEVELOPMENT

CS39A0040 Product & Service Development

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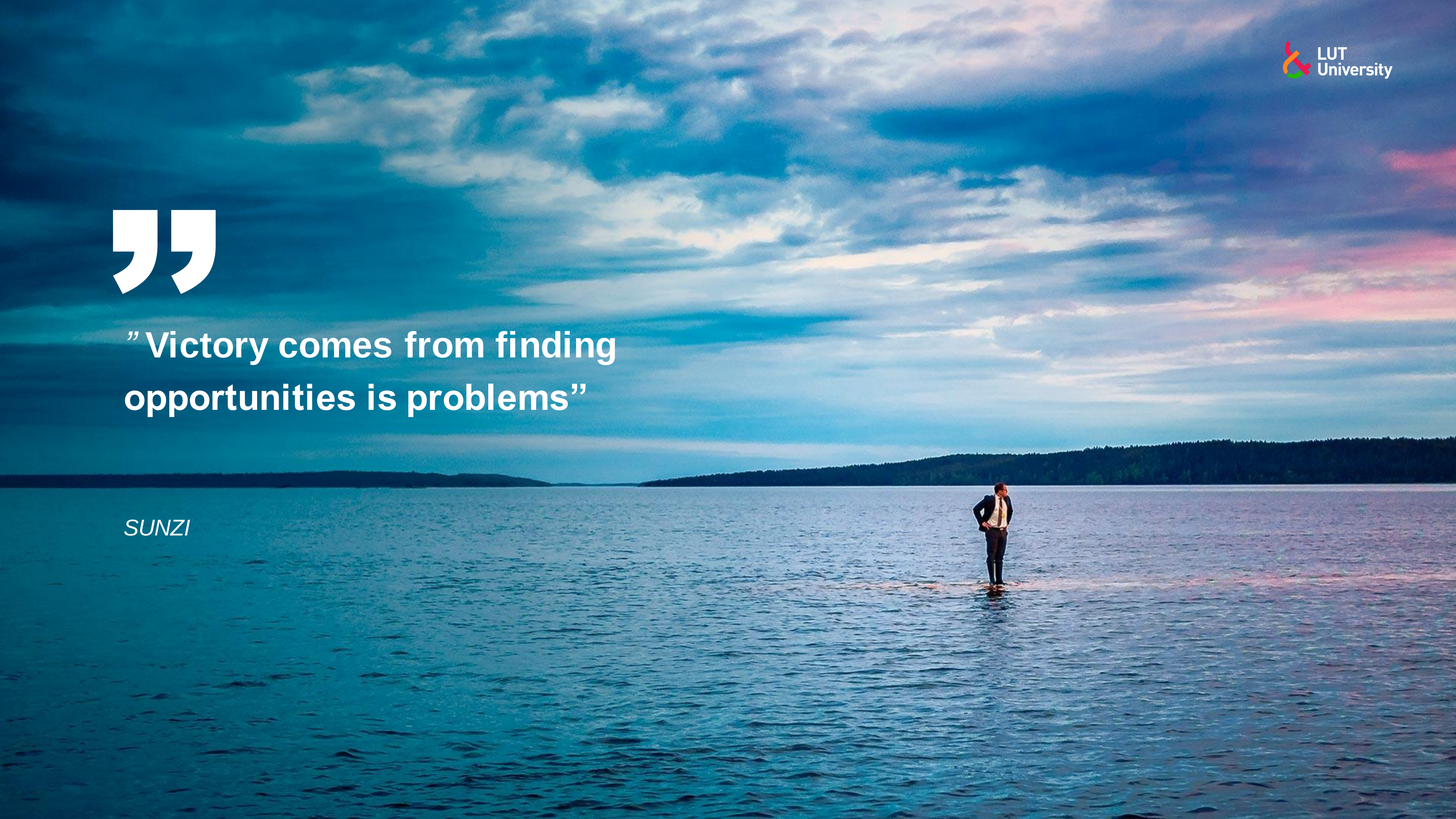
AGENDA FOR LECTURE

- Guest Lecture – Nampuraja Enose, Infosys – Advance Engineering Group
- Lecture 5 – Opportunity Development

”Victory comes from finding opportunities is problems”

SUNZI



OPPORTUNITY DEVELOPMENT - THE BIG PICTURE

Why is opportunity development needed?

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PRODUCT DEVELOPMENT

- **Opportunity Development**
- Concept Development
- Subsystem Development
- System Refinement
- Production Refinement
- Post-Release Refinement

- Requirement Management
- Testing and Verification
- Models & Prototypes
- Design
- Delivery

OPPORTUNITY MATRIX AND PHASE

Methods to define the opportunity and its feasibility

Planning can be formal and informal

OPPORTUNITY : PROBLEM STATEMENT

- The goal of the phase is to develop information characterising the problem to be solved.
- The Opportunity Development phase defines the following areas with the customers & market analysists:
  - Collect market requirements and needs
  - Determining performance measures (Key Performance Indicators)
  - Define requirement-measure correlations
  - Establishing ideal values of performance
  - Initial value proposition and business model mapping.
- Other project management related approvals are secured, for example:
  - Project Plan and scope
  - Budget & resource plan
  - Change & Risk Management process.

PROJECT OBJECTIVE STATEMENT

- The Project Objective Statement captures the key ideas and characteristics:
  - One of the first deliverables of the project
  - A clear & concise articulation of the project and its goals for the team and the customer(s).
  - An overarching & simple mission statement and while simple
  - A way of uniting a team of the development project
  - It documents the mission the development team has agreed to.

Clear: What is done?, When completed?, How much?

Validated: stakeholders & customers

Concise: Short c. 25 words

Unite: for the team to understand the mission

Informed: Based on information learned from customers & stakeholders

OPPORTUNITY DEVELOPMENT

- Information is collected to understand the market needs and conditions, and product requirements:

Market Requirements

Defines the requirements for a desirable product which the market understands & will adopt.

Performance Measures

Characteristics of the product that will be measured by the development team & stakeholders

Requirement-Measure Correlations

Identify the performance measures having significant effect on each market requirements

Ideal Values

Define the values the market prefers for the performance measures without of trade-offs.

OPPORTUNITY DEVELOPMENT OUTCOMES

MARKET REQUIREMENTS

- Market requirements identify key factors that determine product & services desirability.
- For optimum benefit, there is 5 – 20 high level requirements
- Requirement hierarchy and product focused requirements statements
- Requirement prioritisation
- Stakeholder involvement needed

PERFORMANCE MEASURES (KPI)

- Verify the desirability of the proposed product in the market
- Measures should be quantitative
- Qualitative measures can reduce subjectivity
- Measures should be dependent characteristics of the product
- Unit of measures & prioritisation for KPI
- Measures must be clear to stakeholders & market

OPPORTUNITY DEVELOPMENT OUTCOMES

MARKET REQUIREMENTS & PERFORMANCE MEASURES CORRELATIONS

- Measure (KPI) how well the product meets market requirements
  - 1-to-n KPI can verify requirement
  - 1-to-n requirements can verify a KPI
  - Market requirement must have min. 1 KPI
- Matrix captures the correlations

IDEAL VALUES FOR PERFORMANCE MEASURES

- Ideal values of the market needs & expectations
- Often prioritisation trade-offs
- KPI minimum and maximum limits
- Test cases defined & testing is not done
- Mock-ups are built for communication with stakeholders
  - No models & protypes
- Design is limited to sketches to understand market possiblities

OPPORTUNITY DEVELOPMENT OUTCOMES

COMMON TOOLS AND METHODS

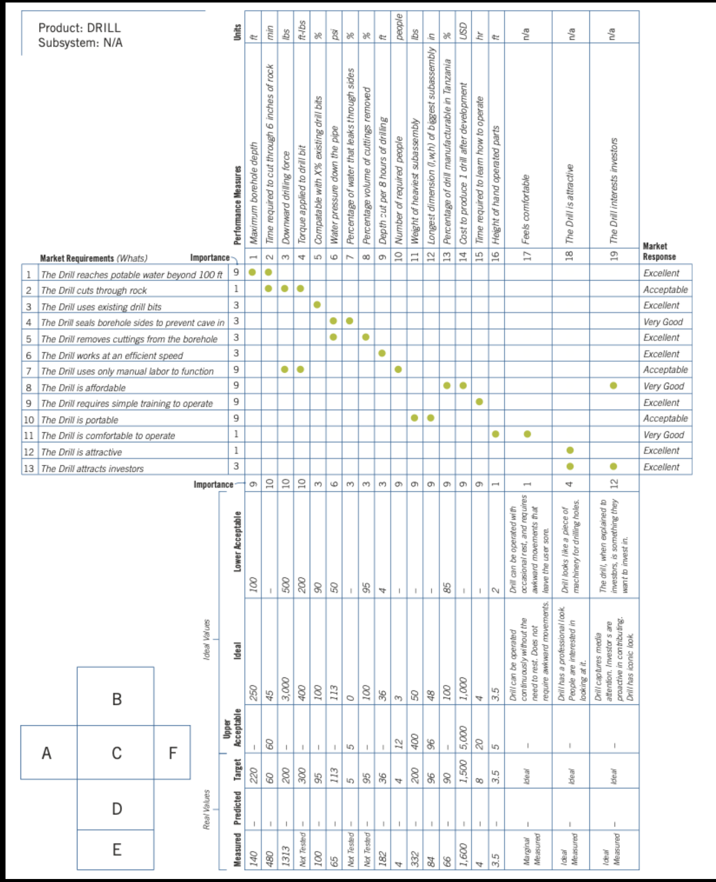
Tools and methods can be divided into the following categories:

- Information collection (e.g. benchmarking, surveys)
- Market requirement identification & prioritisation, e.g.:
  - Requirement Matrix
  - Requirement Hierarchy
- General design work, e.g.:
  - Basic Design Process
  - Project Objective Statement

COMMON MISTAKES (PITFALLS)

- Teams assume and do not validate what the market wants
- Team uses subjective measures and objective KPI
- Refining work before getting feedback. Revise work based on feedback
  - Fail fast and improve
- Opprotunity Phase value is not understood or not done
- Not having a sufficient understanding of the market requirements risk subsequent stages of development.

REQUIREMENT MATRIX



Product: DRILL
Subsystem: N/A

#	Market Requirements (Whats)	Importance	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	Market Response
	Performance Measures		Maximum borehole depth	Time required to cut through 6 inches of rock	Downward drilling force	Torque applied to drill bit	Compatible with X% existing drill bits	Water pressure down the pipe	Percentage of water that leaks through sides	Percentage volume of cuttings removed	Depth cut per 8 hours of drilling	Number of required people	Weight of heaviest subassembly	Longest dimension (l,w,h) of biggest subassembly	Percentage of drill manufacturable in Tanzania	Cost to produce 1 drill after development	Time required to learn how to operate	Height of hand operated parts	Feels comfortable	The Drill is attractive	The Drill interests investors	
	Units		ft	min	lbs	ft-lbs	%	psi	%	%	ft	people	lbs	in	%	USD	hr	ft	n/a	n/a	n/a	
1	The Drill reaches potable water beyond 100 ft	9	●	●																		Excellent
2	The Drill cuts through rock	1		●	●	●																Acceptable
3	The Drill uses existing drill bits	3					●															Excellent
4	The Drill seals borehole sides to prevent cave in	3						●	●													Very Good
5	The Drill removes cuttings from the borehole	3						●		●												Excellent
6	The Drill works at an efficient speed	3									●											Excellent
7	The Drill uses only manual labor to function	9			●	●						●										Acceptable
8	The Drill is affordable	9													●	●					●	Very Good
9	The Drill requires simple training to operate	9															●					Excellent
10	The Drill is portable	9											●	●								Acceptable
11	The Drill is comfortable to operate	1																●	●			Very Good
12	The Drill is attractive	1																		●		Excellent
13	The Drill attracts investors	3																		●	●	Excellent
	Importance		9	10	10	10	3	6	3	3	3	9	9	9	9	9	9	1	1	4	12	
	Ideal Values: Lower Acceptable		100	–	500	200	90	50	–	95	4	–	–	–	85	–	–	2	Drill can be operated with occasional rest, and requires awkward movements that leave the user sore.	Drill looks like a piece of machinery for drilling holes.	The drill, when explained to investors, is something they want to invest in.	
	Ideal Values: Ideal		250	45	3,000	400	100	113	0	100	36	3	50	48	100	1,000	4	3.5	Drill can be operated continuously without the need to rest. Does not require awkward movements.	Drill has a professional look. People are interested in looking at it.	Drill captures media attention. Investor s are proactive in contributing. Drill has iconic look.	
	Ideal Values: Upper Acceptable		–	60	–	–	–	–	5	–	–	12	400	96	–	5,000	20	5	–	–	–	
	Ideal Values: Target		220	60	200	300	95	113	5	95	36	4	200	96	90	1,500	8	3.5	Ideal	Ideal	Ideal	
	Real Values: Predicted		–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	
	Real Values: Measured		140	480	1313	Not Tested	100	65	Not Tested	Not Tested	182	4	332	84	66	1,600	4	3.5	Marginal Measured	Ideal Measured	Ideal Measured	

The Requirement Matrix:
- Captures all information about the product in one place
- Translates market requirements into performance measures
- Revised throughout the project

- Six sections of the matrix:
  A. Market Requirements
  B. Performance Measures & Units
  C. Requirement-Measure Correlations
  D. Ideal Values
  E. Real Values
  F. Market Response.

Refer to book Section 11.55

- Clear Statement of market & engineering requirements that collects markets needs

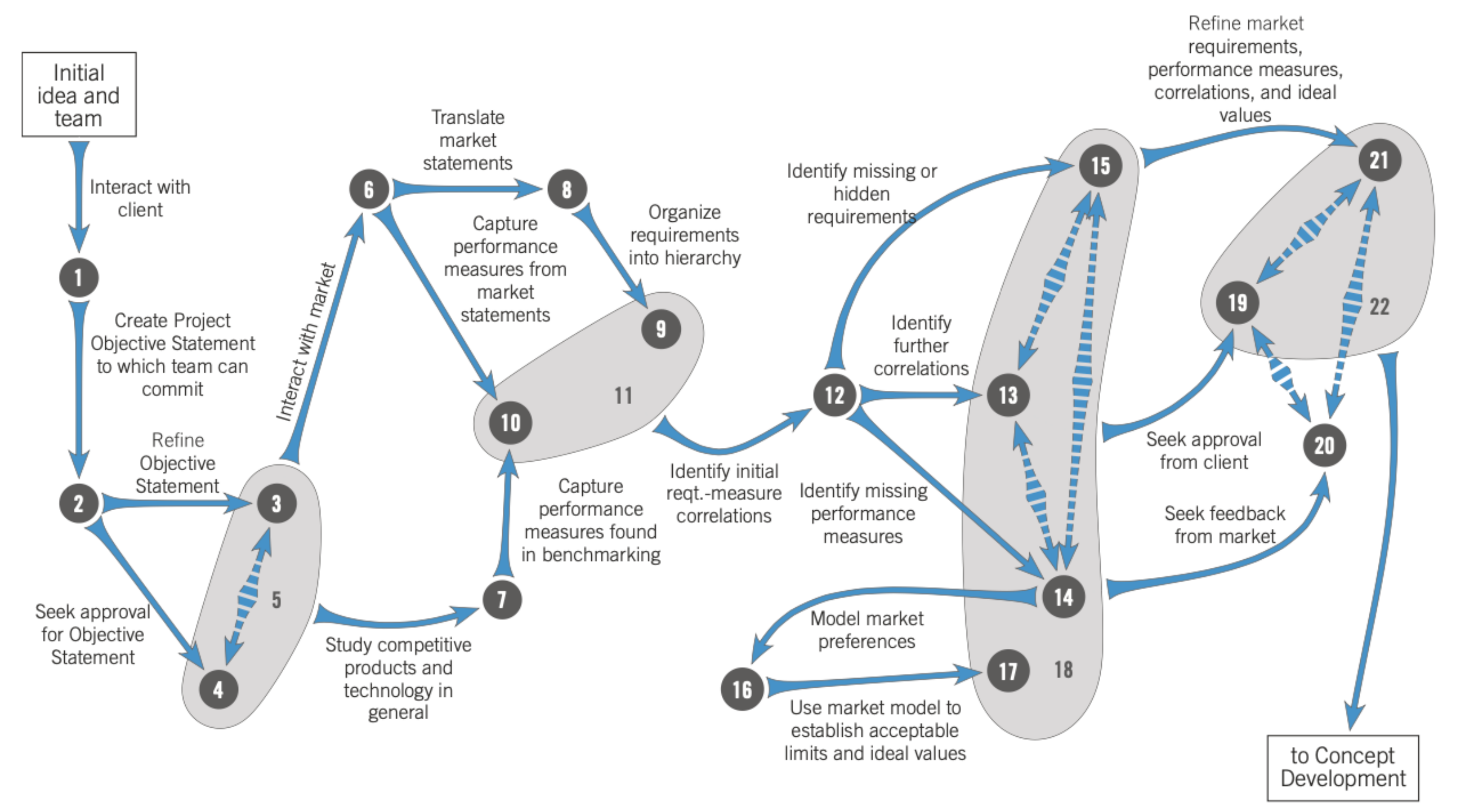
OPPORTUNITY DEVELOPMENT PHASE

	Requirement Information	Artifacts	Checking Criteria	Approval Criteria
Requirements	**Market Requirements**	Matrix Section A	Consistent level of generality? Capture the most important requirements? Appropriate number of requirements? Reasonable differences in importance?	Complete & appropriate as evaluated by market representative
	Performance Measures	Matrix Section B	Clearly measurable? Capture market requirements well? Generally dependent, rather than independent? Units given & appropriate? Number appropriate? Appropriate importance ratings?	Market representatives (for less technical measures) and/or project approvers (for highly technical measures) find measures to be appropriate.
	Requirement-Measure Correlations	Matrix Section C	All requirements have min. 1 measure? All measures have min. 1 requirement? More than just 1-2-1 correlations? Appropriate, defensible correlations?	Judged appropriate by project approvers
	Ideal Values	Matrix Section D	Values make sense? Values are consistent with market requirements?	Judged appropriate by project approvers and market representatives
Tests	Not Required	Reports of team interactions with market representatives (e.g., surveys, interviews, etc.)	Are the interactions accurately conveyed in the report?	Not approved directly, but used to support approval of the requirements
Models	Not Required	Simple models that relate desirability to measured performance of competitors (e.g., screen size, battery life)	Do the models make logical sense?	Not approved directly, but used to support approval of the requirements
Prototype	Not Required	Rough prototypes (foamboard, clay, cardboard, plywood) used to communicate with market representatives.	Do the prototypes facilitate communication with market representatives?	Not approved directly, but used to support approval of the requirements
Design	Not Required	Rough sketches or drawings of competitors or generic product possibilities.	Do the sketches or drawings facilitate communication with market representatives?	Not approved directly, but used to support approval of the requirements
	Tools:	Basic design process, competitive benchmarking, financial analysis, focus groups, interviews, observational studies, patent searches, planning canvas, project objective statement, quality function deployment, requirements matrix, surveys.		
	Common Pitfalls:	Assuming, not validating; using only subjective performance measures; delaying feedback; devaluing the opportunity development stage; spending too much time.		

PRODUCT-FOCUSED REQUIREMENT STATEMENTS

Guideline	Customer Statement	Good Requirement Statement	Bad Requirement Statement
Express the requirement with positive, not negative, phrasing	It just doesn't have the torque of a corded saw and will choke in a slight bind while cutting plywood sheets.	The saw resists binding when cutting plywood.	The saw doesn't bind when cutting plywood.
Express the requirement as specifically as the user statement	I wish it had a case to protect it from dents and dings	The saw is resistant to dents and dings.	The saw is rugged.
Express the requirement as a requirement of the product, not the environment or the user	The trigger safety interlock is aggravating! I have incredibly large hands and still find the trigger interlock a stretch to reach.	The saw can be started easily by most people, even with safety interlock features active.	People can easily reach the safety interlock.
Express a requirement, rather than a performance measure	The battery lasts for three hours of typical carpentry work.	The battery lasts a long time. *or* The battery life allows typical carpentry work.	The battery lasts for three hours.
Express a requirement, rather than a product feature	Having the blade on the left side gives a right-handed user a great line of sight.	The saw provides clear sight to the cut.	The saw has the blade on the left side.
Express the requirement independent from its importance	The saw does not have a light which I miss.	The saw illuminates the cut area.	The saw should illuminate the cut area. *or* The saw must illuminate the cut area.

TOP LEVEL ACTIVITY MAP



OUTCOMES;

1. Client's wishes about scope, schedule, 8 resources
2. Draft Project Objective Statement
3. Refined Project Objective Statement
4. Approved Project Objective Statement
5. Market & customer statements about the product
6. General benchmarking data
7. Product-focused requirement statements
8. Initial market requirements
9. Initial performance measures
10. Initial measures and requirements
11. Initial requirement-measure correlations
12. Refined requirement-measure correlations
13. Refined performance measures
14. Refined market requirements
15. Models of market acceptance vs performance
16. Ideal values and acceptable limits for performance measures
17. Initial Opportunity Definition
18. Approval of opportunity development
19. Market feedback on requirements, measures, and ideal values
20. Refined Opportunity Definition
21. APPROVED OPPORTUNITY DEFINITION


ANY
QUESTIONS
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